Train / Calibration Phase

Train/Validation set

$\mathcal{D}_{\text{train/val}} = \{(X_i, Z_i, Y_i)\}$

Optimized by quantile loss

Quantile Model

create lower & upper bound

$[\hat{\ell}(X_i), \hat{u}(X_i)]$

Calibration set

$\mathcal{D}_{\text{cal}} = \{(X_i, Z_i, Y_i)\}$

sample i : $\{(X_i, Z_i, Y_i)\}$

Nonconformity Score

$s(X_i, Z_i) =$

$\max\left\{\hat{\ell}(X_i) - Z_i, Z_i - \hat{u}(x_i)\right\}$

Group by true class

Group calibration samples by true ordinal class k

$I_k = \{i : Y_i = k\}$

Class specific calibration score

Non-conformity scores for each class

$S_{k,i} = s(X_i, Z_i) \qquad i \in I_k$

Class-specific Conformal Corrections

$q_k = \widetilde{S}_{k,(r_k)}$: r_k-th order statistic of the class-k calibration scores

X_i : input, Z_i : numeric target, Y_i : ordinal class

Test Phase

Test input x

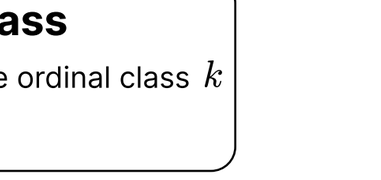
Trained Model Outputs

$[\hat{\ell}(x),\ \hat{u}(x)]$

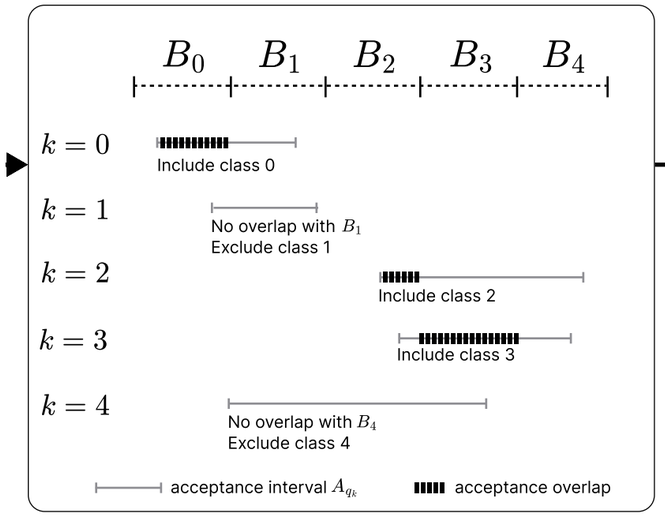
Candidate-wise evaluation

$A_{q_k}(x) = \{z \in \mathbb{R} : s(x, z) \le q_k\}$

candidate acceptance set for class k



q_k may expand or shrink the base interval.



Raw ordinal set

Example raw set

$C_{raw}(x) = \{0, 2, 3\}$

Is empty?

Yes

Empty-set Fallback

$C_{fallback}(x) = \{0, 1, 2, 3, 4\}$

No

Ordinal Hull

$C_{hull}(x) = \{0, 1, 2, 3\}$

Final contiguous set

Example final set

$C(x) = \{0, 1, 2, 3\}$

$S_{k,i}$: class-k calibration score, q_k : class-specific correction, B_k : numeric bin for class k